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Report OF 4 Bit binary adder / subtractor

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4 Bit binary adder and subtractor

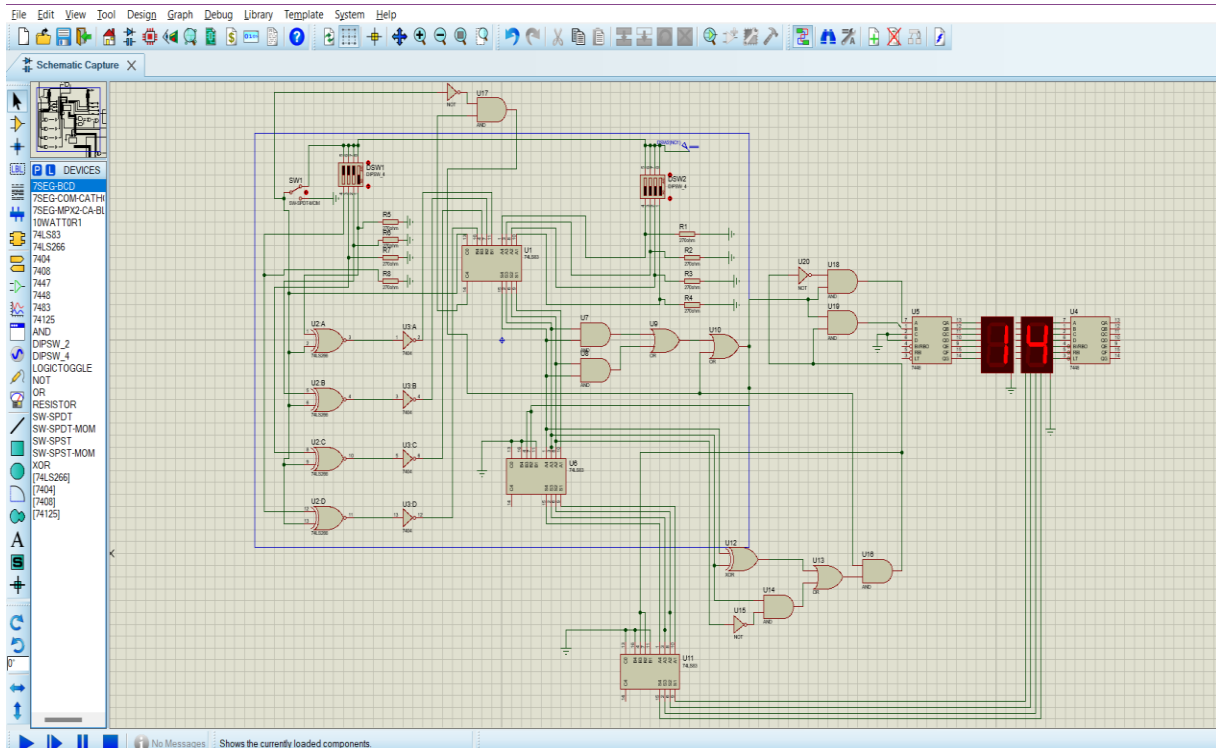
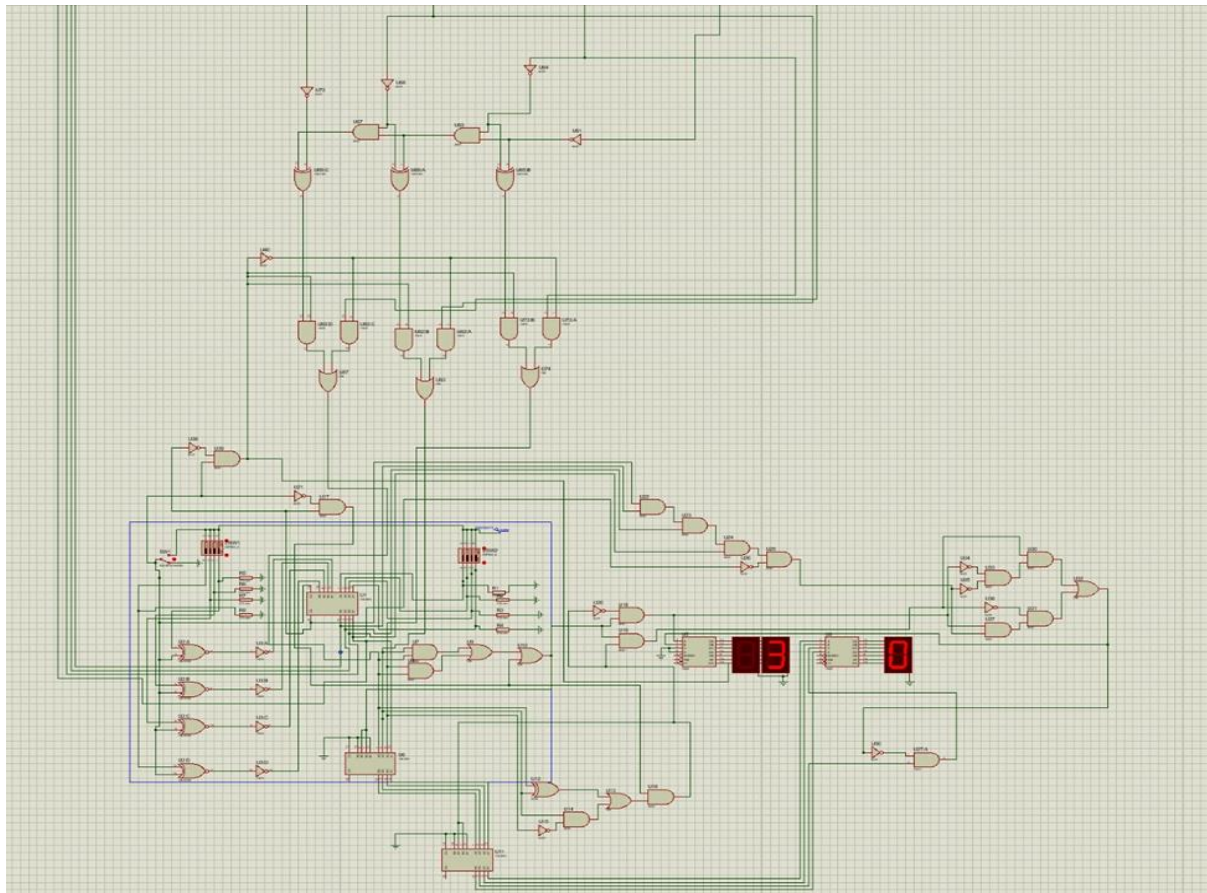
Abstract

This report aims to describe a binary adder circuit that can perform addition of two 4-bit binary numbers. The circuit uses various components such as 7483 4-bit adder ICs, 74138 decoder IC, and 7432 OR gate IC. The report will detail the functions of each component and how they

How the Circuit Works

- **Input Processing:** The 4-bit binary numbers (A and B) are fed into the circuit.
- **Full Adders in Action:** Four full adders are cascaded, meaning the carry-out from one adder becomes the carry-in for the next.
- **Leveraging K for Subtraction:** Here's the key trick: When $K = 1$ (subtraction), the B input for the first full adder is Xored with the control signal (K). This essentially creates the 2's complement of B.
- **2's Complement:** A way to represent negative numbers in binary. Inverting the bits of B and adding 1 performs subtraction. The XOR operation with K achieves this efficiently.
- **Full Adder Magic:**
 - If $K = 0$ (addition): The full adders simply perform regular binary addition ($A + B + \text{Carry-in}$) on each bit position.
 - If $K = 1$ (subtraction): The full adders effectively perform ($A + B' + 1 + \text{Carry-in}$) due to the 2's complement trick with the XOR operation. This achieves subtraction.
- **Output :** The final sum (S) from the last full adder represents the result of the addition or subtraction. The carry-out from the last adder might also be relevant depending on the specific operation.

The protues design of the circuit



Functions of Each Component

Adder(7483)

- **Convenience:** Integrates all necessary components for 4-bit addition/subtraction.
- **Efficiency:** Performs the chosen operation on all 4 bits simultaneously

XOR(7486)

4-bit adder/subtractor relies on XOR gates for control and 2's complement generation within the full adders to achieve subtraction functionality.

AND(7408)

is used along with XOR gates to determine the carry output (C) based on the input conditions

OR(7432)

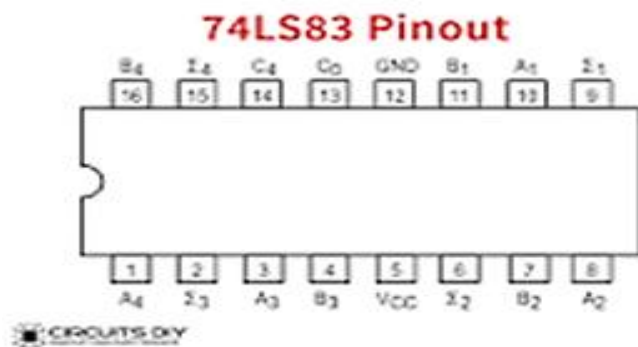
- **Overflow Detection:** Identifying situations where the addition result exceeds the representable range (e.g., adding two positive numbers with the highest possible value).
- **Sign Extension:** Propagating the sign bit in specific scenarios, which can be important for interpreting the result as positive or negative

Decoder(7448)

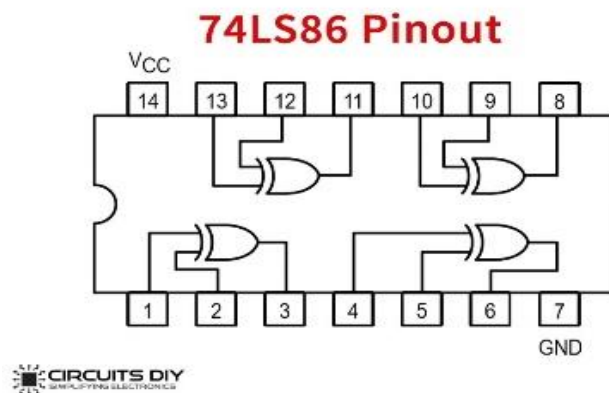
is used to decode the input binary numbers and select the appropriate 7483 IC for each bit of the input. The IC has three inputs, A, B, and C, which represent the three most significant bits of the input binary numbers. The output of the IC is used to select the appropriate 7483 IC for each bit of the input

The block diagram

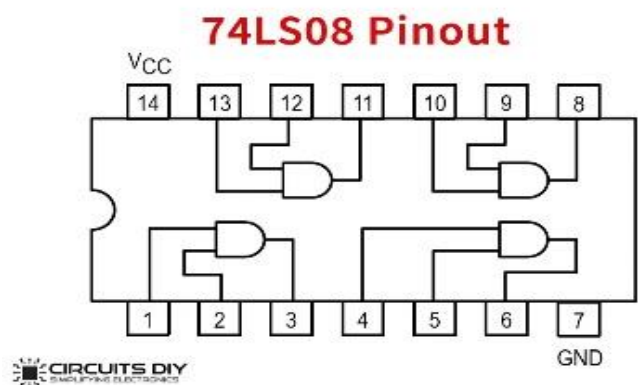
1-Adder



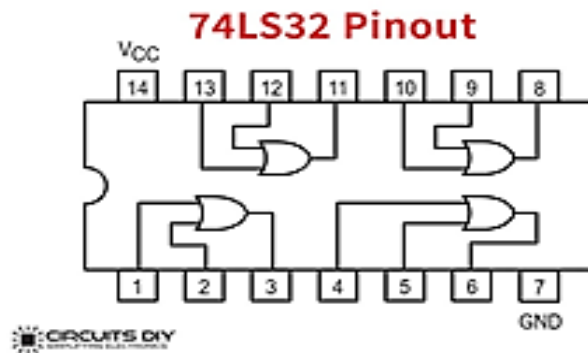
2-XOR



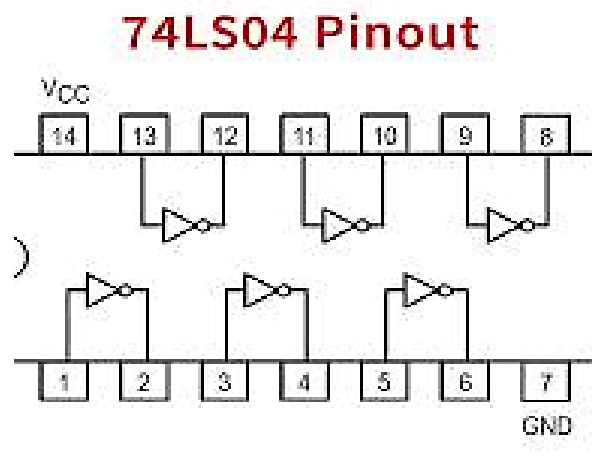
3-AND



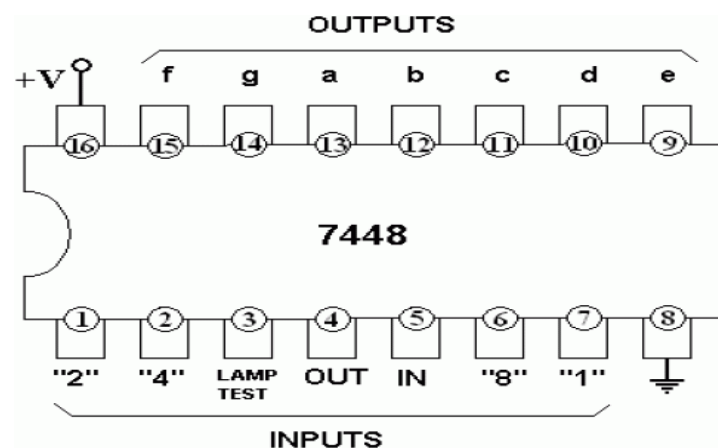
4-OR



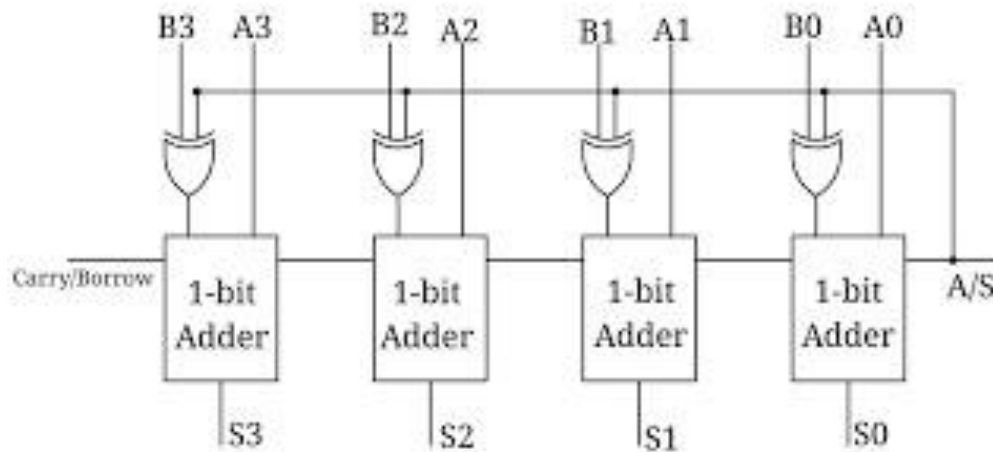
5-NOT



6-DECODER



Circuit Diagram



Advantages

- The circuit design is simple, reliable, and efficient, making it ideal for understanding fundamental digital arithmetic operations.
- It is entirely hardware-based and does not require programming, which enhances its suitability for hands-on learning.
- The circuit operates with low power consumption and fits easily on compact breadboards or small-scale PCBs.
- The modular structure allows easy scalability to higher-bit operations, such as 8-bit or 16-bit arithmetic.

Limitations

- The design is restricted to 4-bit binary numbers, which limits its use in real-world applications requiring larger data widths.
- It lacks overflow detection, which may result in incorrect outputs when the result exceeds the 4-bit range.

- The circuit does not account for signed number operations unless additional logic is implemented.
 - Inputs are provided manually through switches, which may not be practical for continuous or automated use.
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Applications

- Used in the design of simple Arithmetic Logic Units (ALUs) found in digital processors.
 - Serves as an educational tool in digital electronics and logic design laboratories.
 - Can be integrated into small embedded systems for basic arithmetic operations.
 - Suitable for use in logic circuit testing and functional verification setups.
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Conclusion

The 4-bit binary adder and subtractor project effectively demonstrates core principles of digital computation. It highlights the practical application of logic gates and binary arithmetic through a hands-on hardware implementation. While limited in scope, the project provides a strong foundation for understanding how larger digital systems perform arithmetic operations. It is an excellent exercise in circuit design, logic comprehension, and hardware implementation for educational and prototyping purposes.